

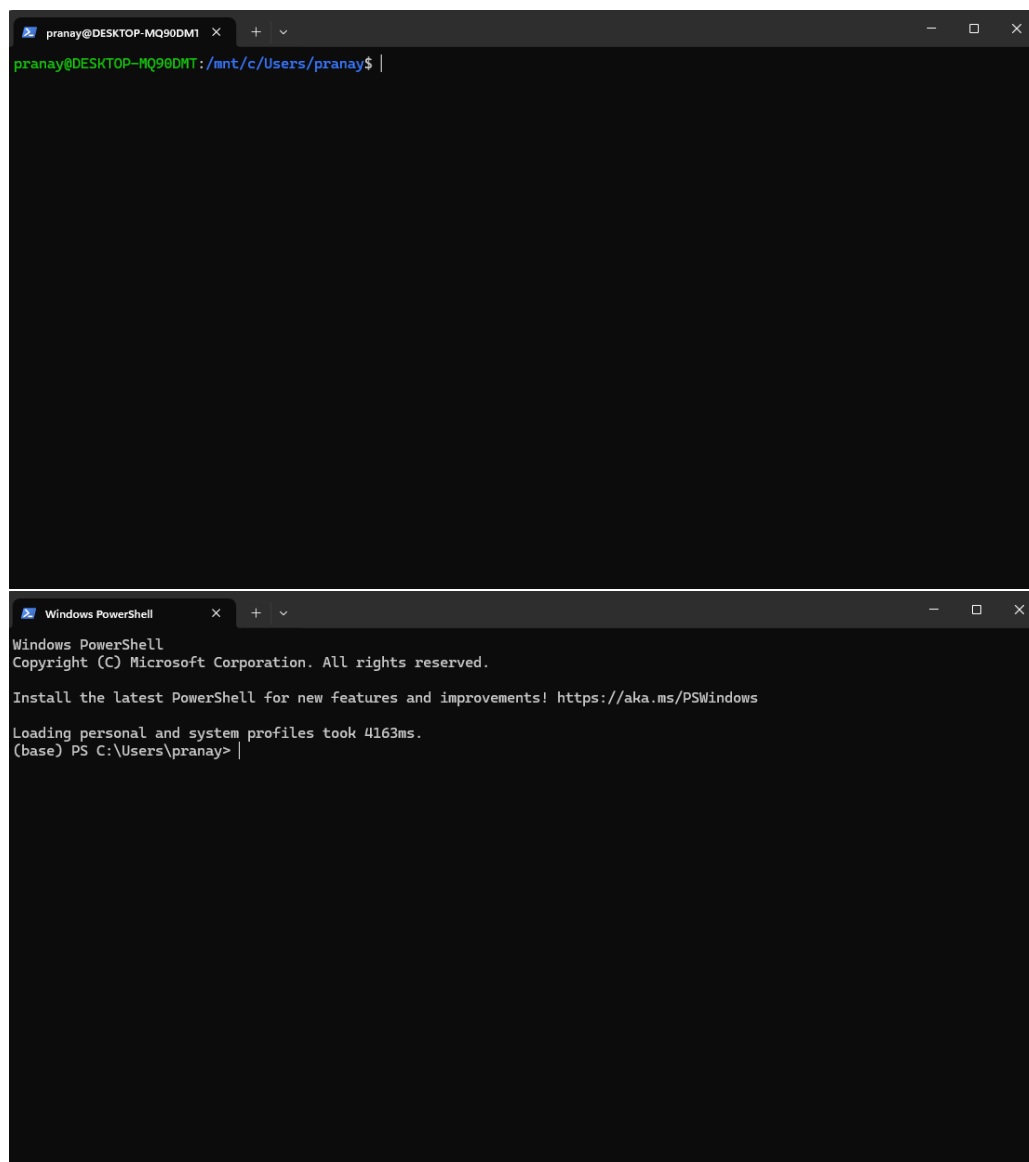
A Quick Handout for Basic UNIX Commands

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Introduction

The UNIX shell is a powerful command-line interface for interacting with your computer. This handout provides a reference for some of the most common and essential commands for file system navigation and manipulation. It also explains how to learn more about any command using the built-in manual pages.



```
pranay@DESKTOP-MQ90DM1 x + v
pranay@DESKTOP-MQ90DM1: /mnt/c/Users/pranay$ |

Windows PowerShell x + v
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Loading personal and system profiles took 4163ms.
(base) PS C:\Users\pranay> |
```

Accessing the Server

Before you can use any of the commands below on the lab server, you must connect via SSH. Follow these steps:

1. **Open your terminal or Command Prompt.**

- On Windows, you can use *PowerShell* or *Command Prompt*.
- On macOS or Linux, open the *Terminal* application.

2. **Connect via SSH:** Type the following, replacing `student_adm_number` with your own admission number (all lowercase):

```
ssh student_adm_number@172.16.201.16
```

3. **Authenticate:** When prompted, enter the default password:

Password: `cse@123`

4. **Change your password:** On first login you will be forced to update your password. At the prompt, type:

Then follow the on-screen instructions:

- Enter your current password (`cse@123`).
- Choose a new password and type it twice when prompted.
- Make sure your new password is memorable but secure.

5. **Always remember your password!** You will need it each time you connect.

Core File System Commands

Here is a list of fundamental commands. In the examples, text inside `<...>` is a placeholder you should replace with your own file or directory name.

ls (List)

Lists the files and directories in the current location.

- `ls` — Lists contents of the current directory.
- `ls -l` — Long format (permissions, owner, size, date).
- `ls -a` — Includes hidden files (dot-files).

cd (Change Directory)

Navigates into a different directory.

- `cd <directory_name>` — Move into the specified directory.
- `cd ..` — Up one level to the parent directory.
- `cd ~` or `cd` — Go to your home directory.

pwd (Print Working Directory)

Shows the full path of your current location.

- `pwd`

mkdir (Make Directory)

Creates a new directory.

- `mkdir <new_directory_name>`

cp (Copy)

Copies files or directories.

- `cp <source_file> <destination_file>` — Copy a file.
- `cp -r <source_directory> <destination_directory>` — Recursively copy a directory.

mv (Move)

Moves or renames files and directories.

- `mv <old_name> <new_name>` — Rename.
- `mv <source_file> <destination_directory>` — Move into a directory.

rm (Remove)

Deletes files or directories. **Use with caution!**

- `rm <file_name>` — Delete a file.
- `rm -r <directory_name>` — Recursively delete a directory.

cat (Concatenate)

Displays file contents.

- `cat <file_name.txt>`

Reading Command Documentation with **man**

Every standard UNIX command includes a “manual page” (or *man* page), which is your best source of details.

Accessing a Man Page

Run:

```
man <command_name>
```

For example, `man ls` shows the manual for `ls`.

Structure of a Man Page

Typical sections include:

NAME One-line summary of the command.

SYNOPSIS Syntax: command, options, arguments.

- **Bold** = literal text.
- *Italics* = replace with your own argument.
- [] = optional items.

DESCRIPTION Detailed explanation.

OPTIONS List of flags and their meanings.

EXAMPLES Practical usage examples.

SEE ALSO Related commands/topics.

Navigating in less

Most man pages open in less. Key commands:

- **Down / Up:** ↑ | ↓ or j/k
- **Page Down / Up:** Space / b
- **Search:** /pattern, then n for next
- **Quit:** q

When in doubt, **read the manual!**